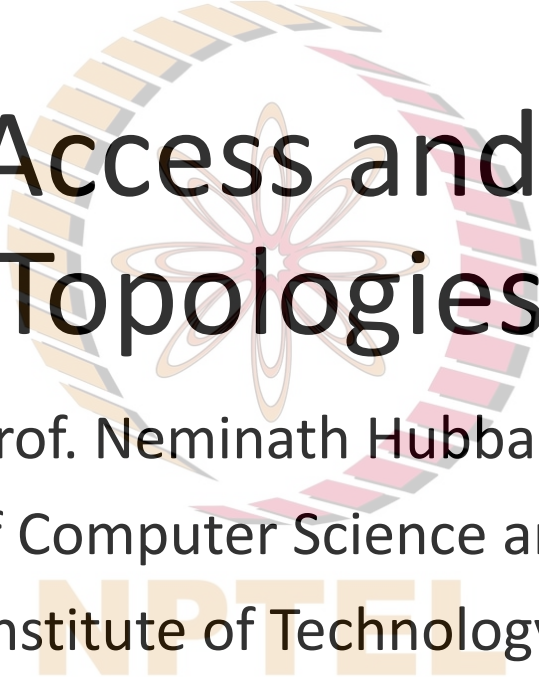


Internet Access and Network Topologies

A large, semi-transparent watermark of the NPTEL logo is centered behind the title. The logo consists of a circular emblem with a stylized flower or star shape in the center, surrounded by a ring of colored segments. Below the emblem, the word "NPTEL" is written in large, bold, orange capital letters.

Prof. Neminath Hubballi

Department of Computer Science and Engineering

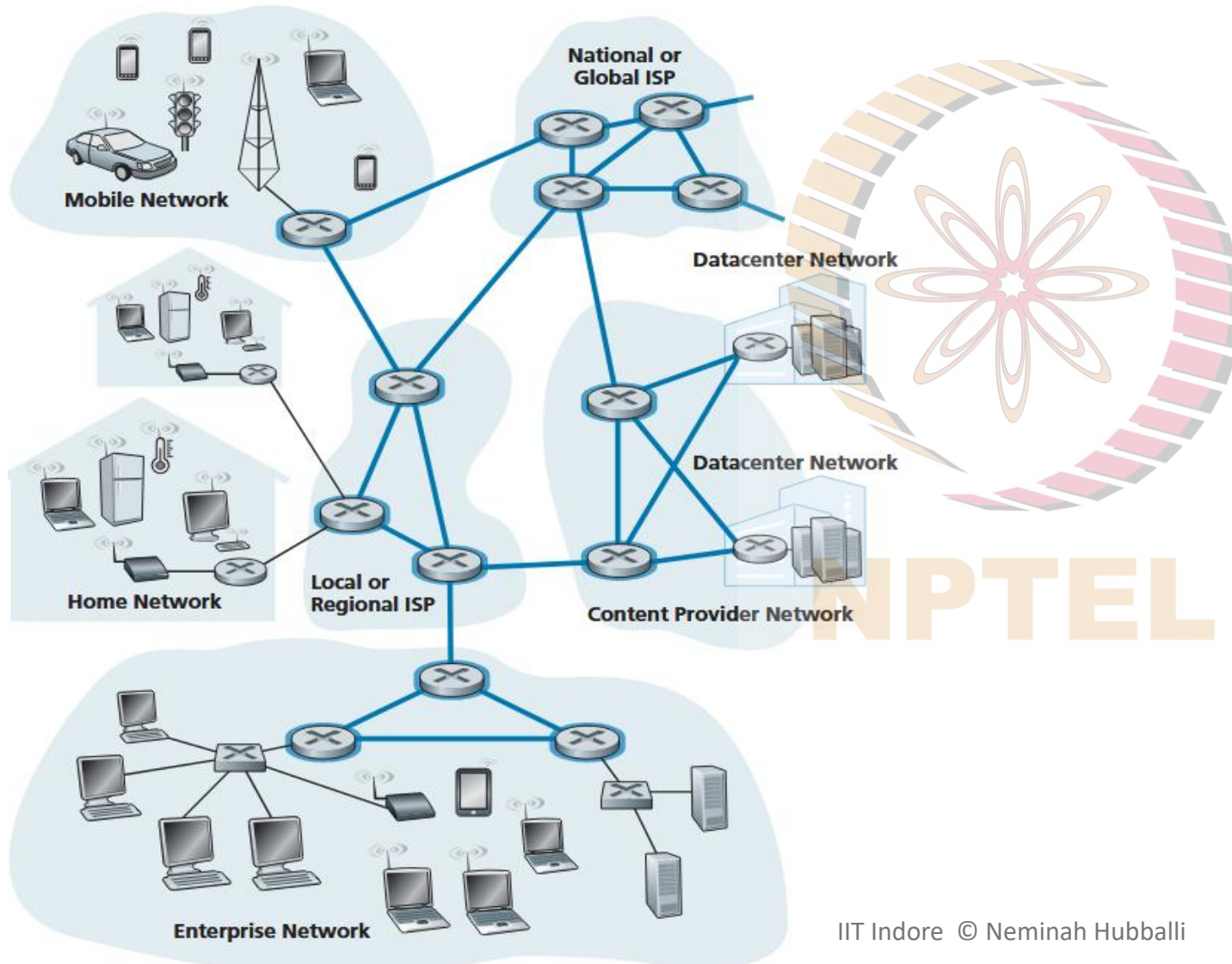
Indian Institute of Technology Indore

Outline

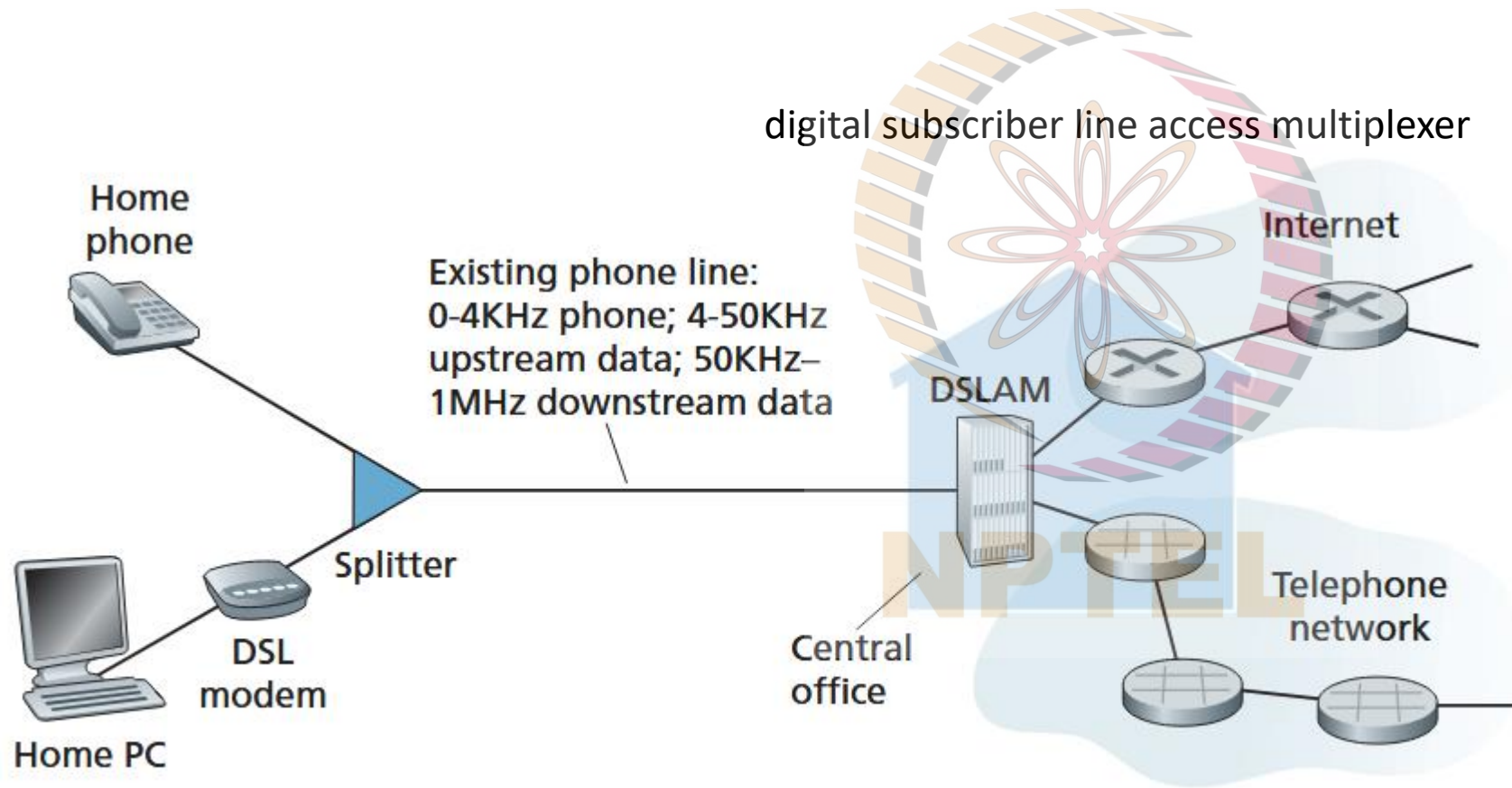
- ☐ Network Edge
 - ☐ Internet access
 - ☐ Physical media
- ☐ Network Core
 - ☐ Physical media
- ☐ Network Design Possibilities
 - ☐ Topologies



Network

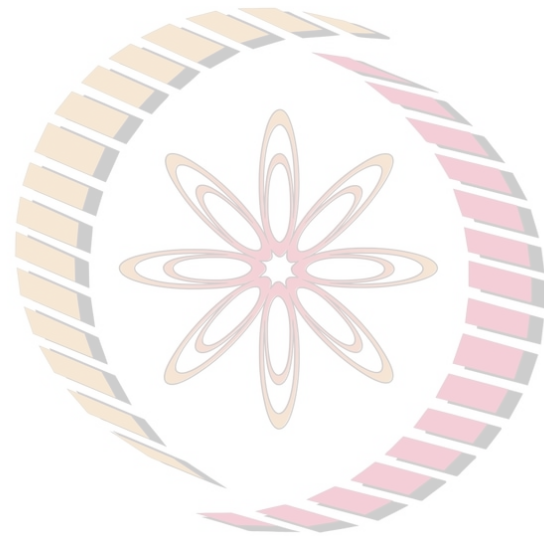


Access Network: DSL and ADSL



52 Mbps downstream
3.5 Mbps upstream





NPTEL

Access Network: DSL and ADSL

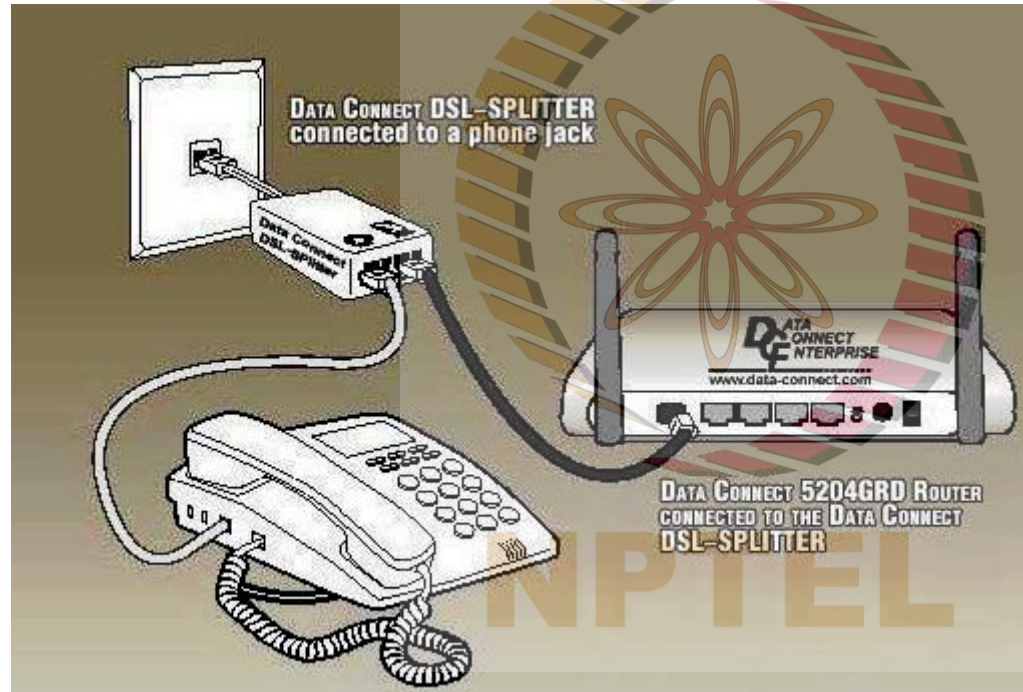
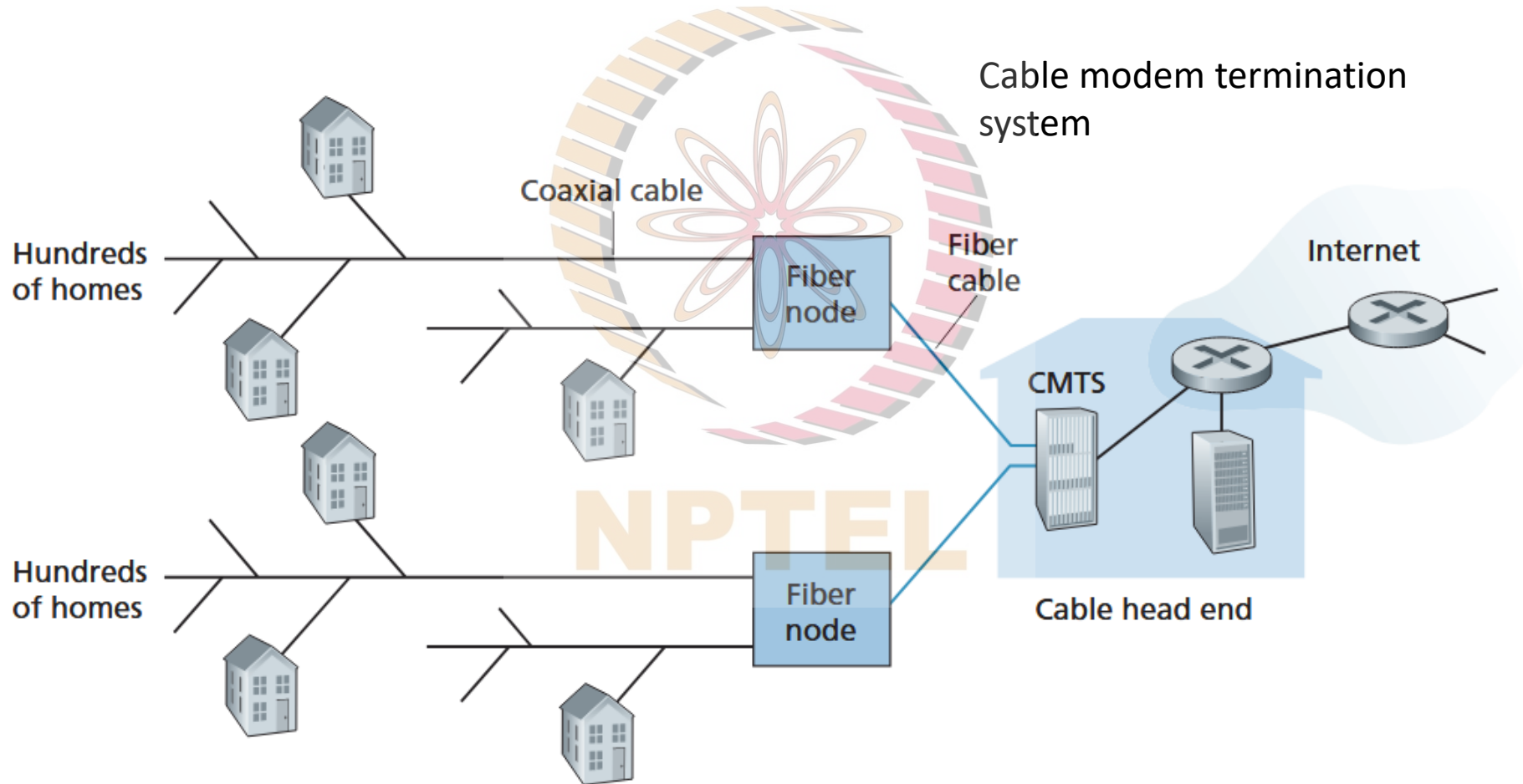
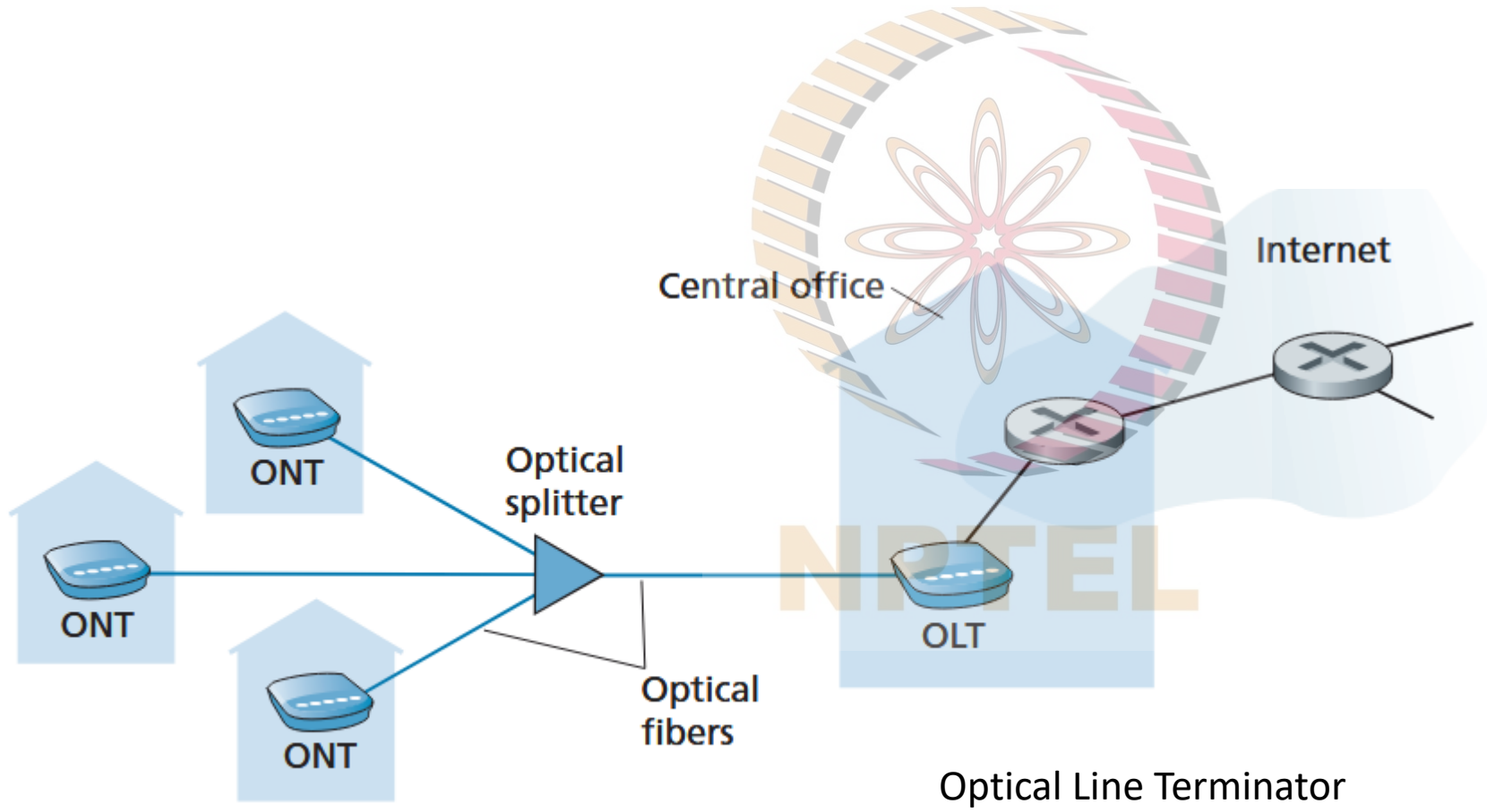


Image courtesy: <https://dataconnectus.com>

Access Network: Cable Internet Access



Access Network: FTTH



Optical Network Terminal

Access Network: FTTH

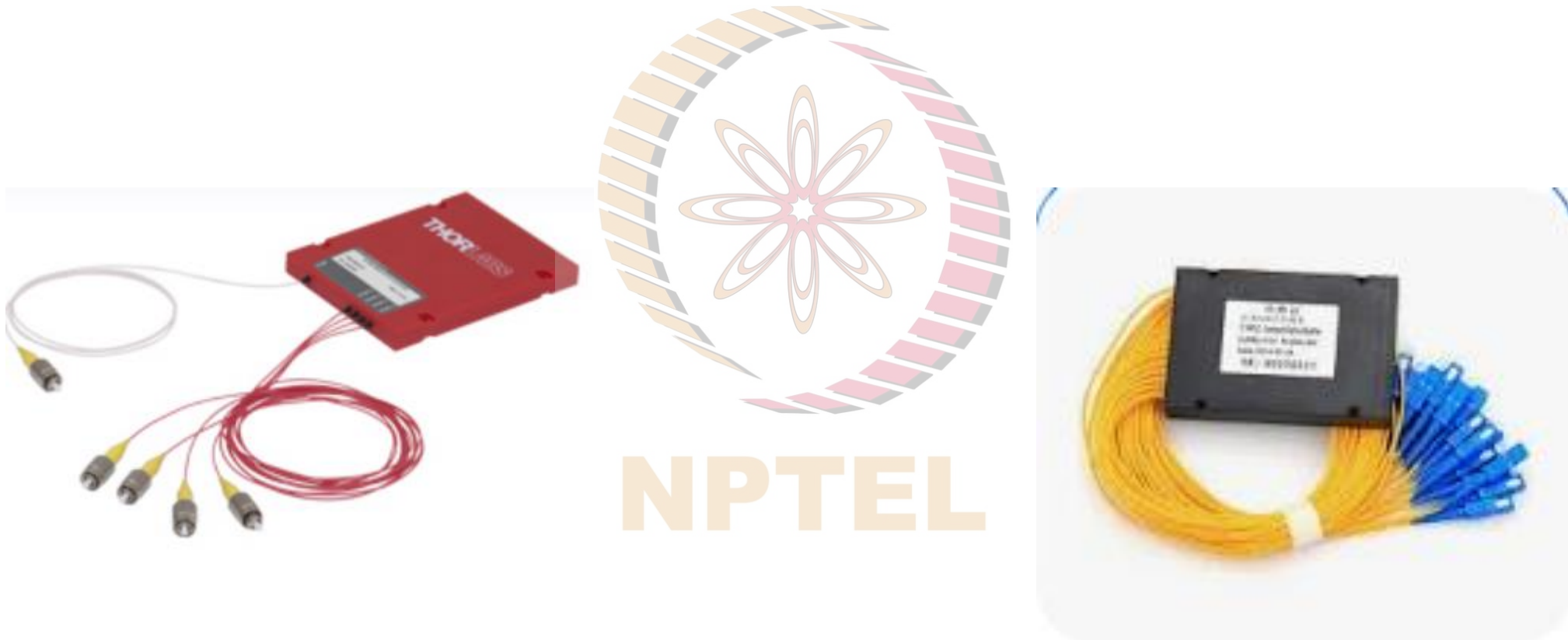


Image Courtesy: <https://techcommerce.in/>

Access Network: FTTH

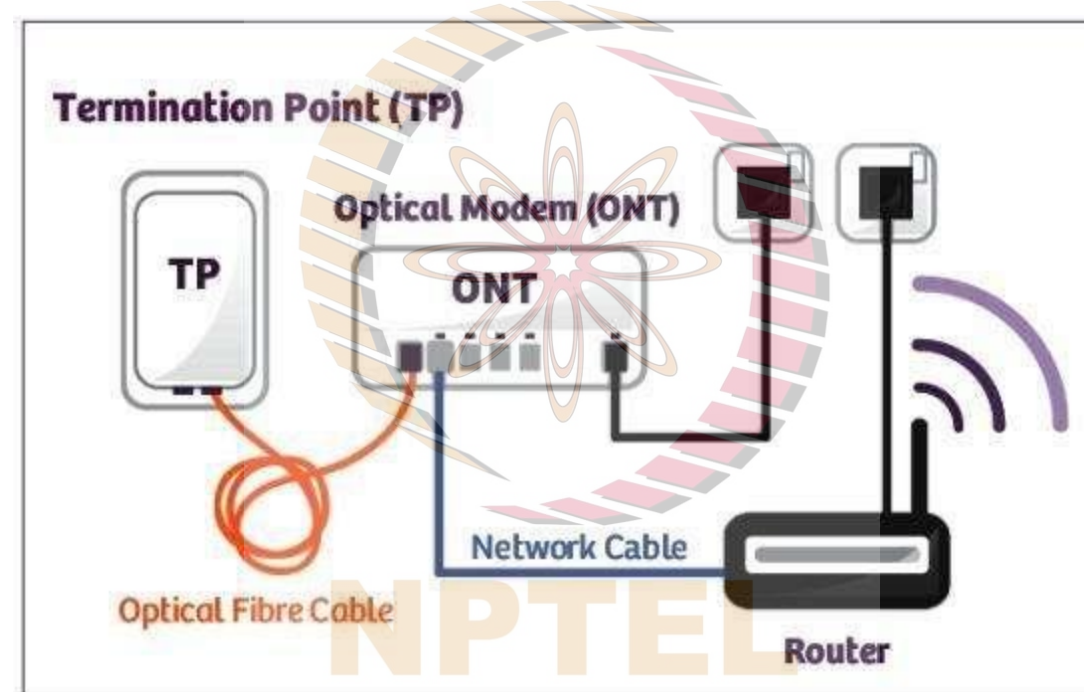
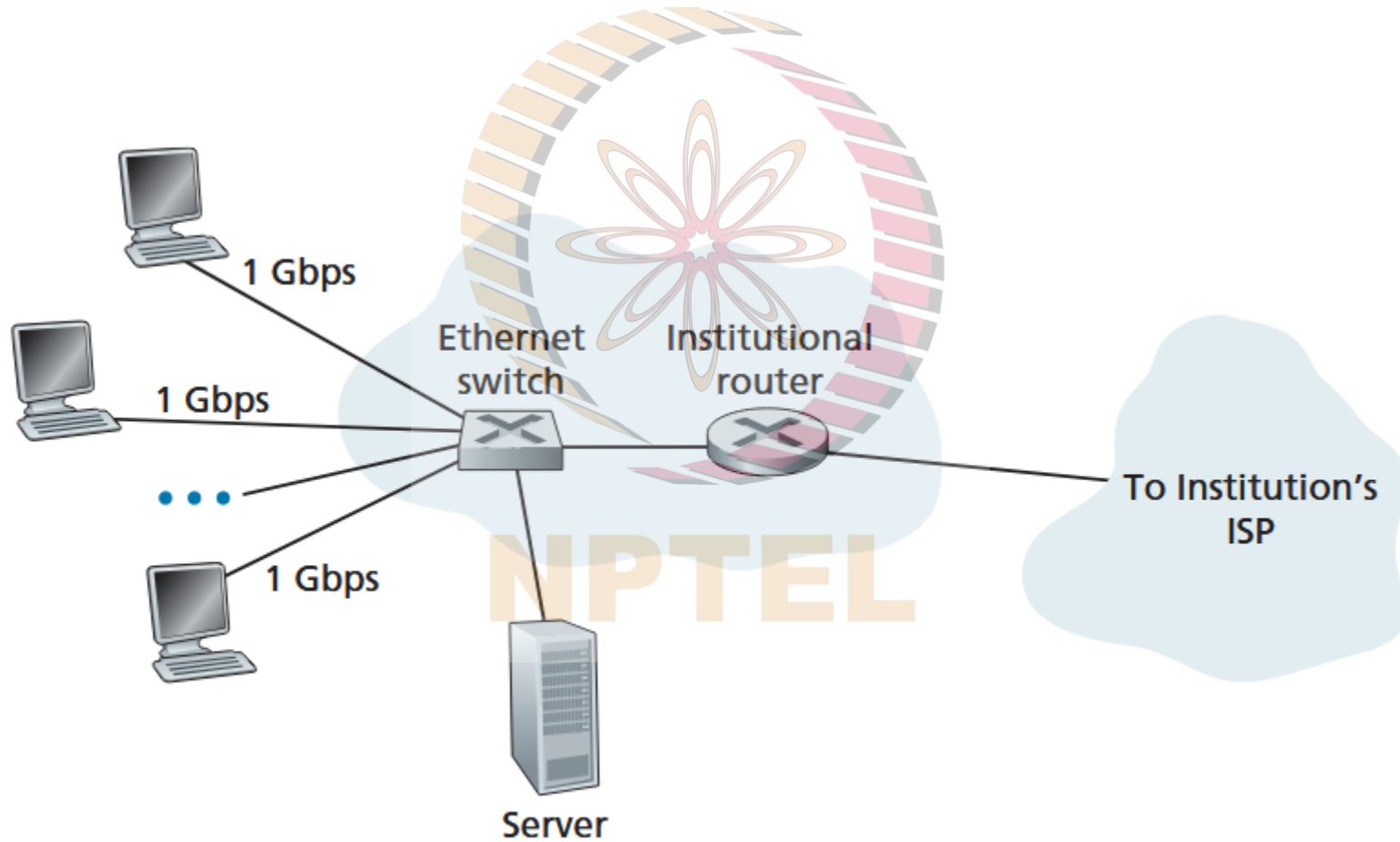


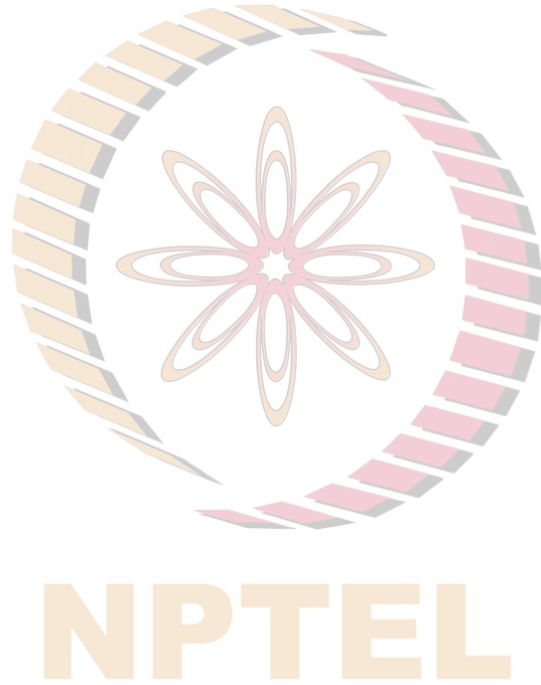
Image Courtesy: Google images

Access Network: Enterprise Users



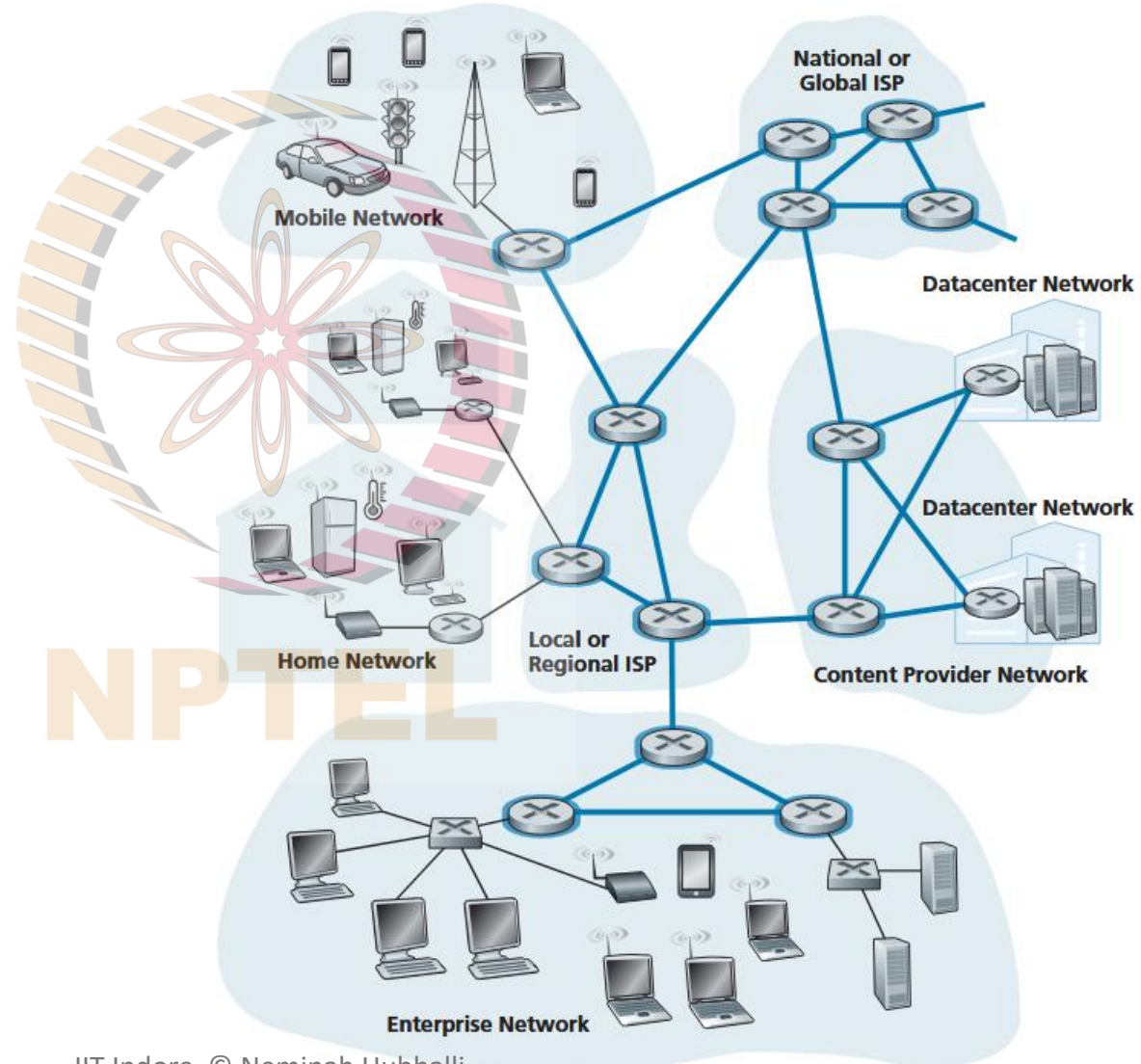
Physical Media

- ☐ Transmission Media
 - ☐ Guided
 - ☐ Unguided



The core of Internet

- ❑ Optical cables



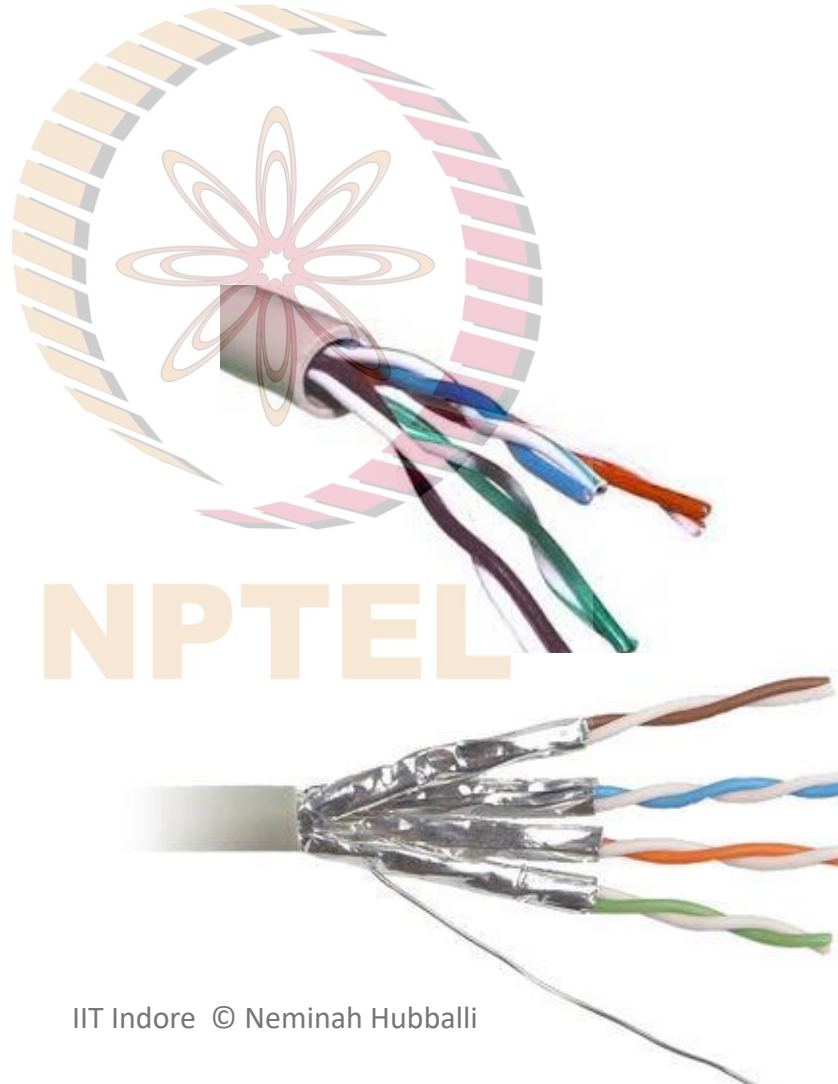
How fast is the backbone ?

Link Type	Speed
OC-1	51.84 Mbit/s
OC-3	155.52 Mbit/s
OC-12	622.08 Mbit/s
OC-24	1244.16 Mbit/s
OC-48	2488.32 Mbit/s
OC-192	9953.28 Mbit/s
OC-768	39,813.12 Mbit/s
OC-1536	79626 Mbit/s
OC-3072	159.252 Gbps
OC-12288	639.00 Gbps

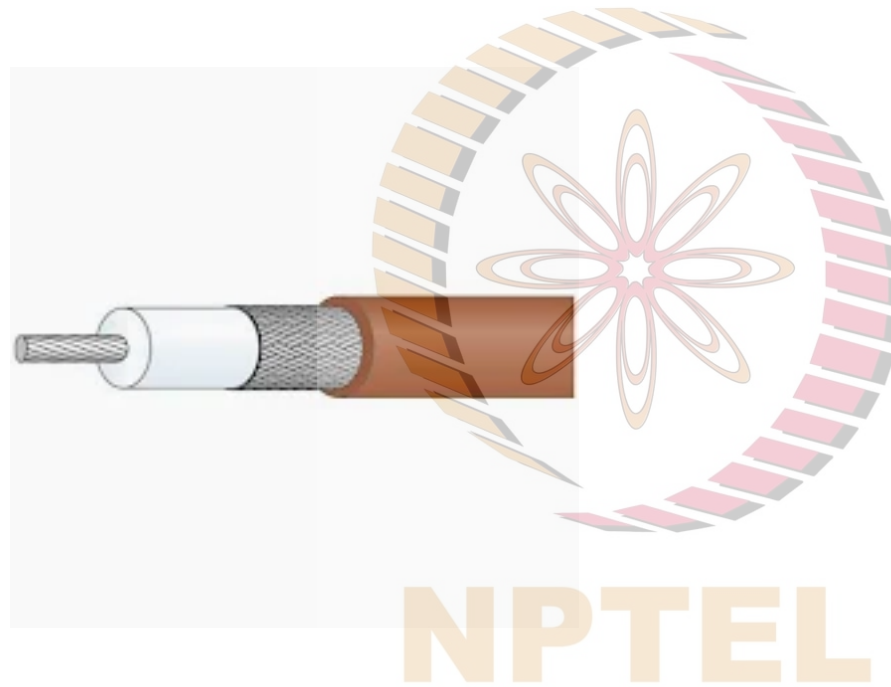
Physical Media: Twisted Pair Cable

☐ Unshielded twisted pair

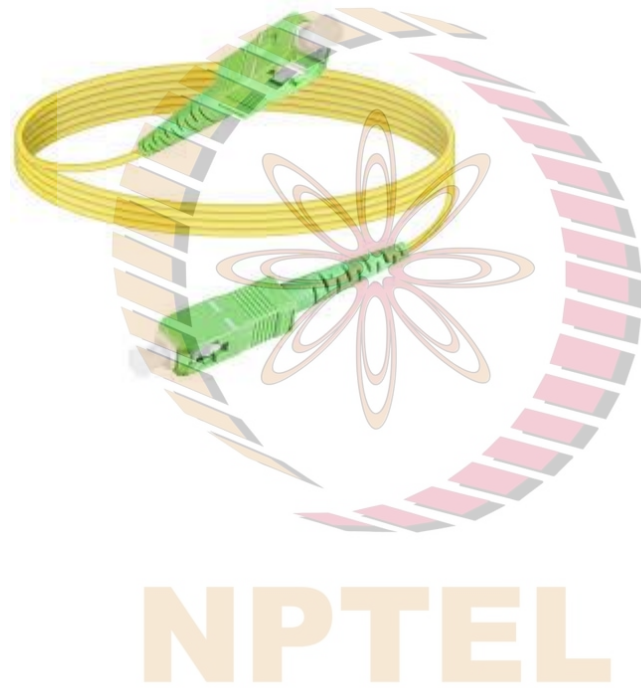
☐ Shielded twisted pair



Physical Media: Coaxial cable



Physical Media: Fiber optic cable



Physical Media: Terrestrial Radio Channels

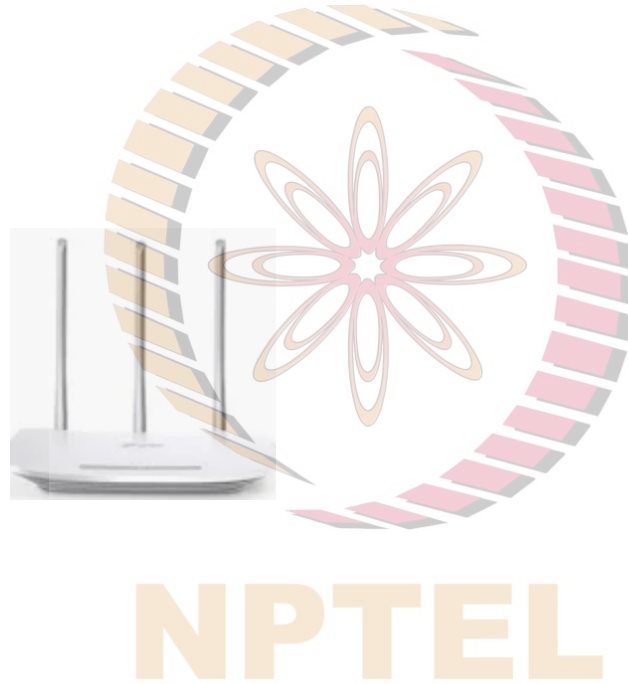


Image Courtesy: Google images